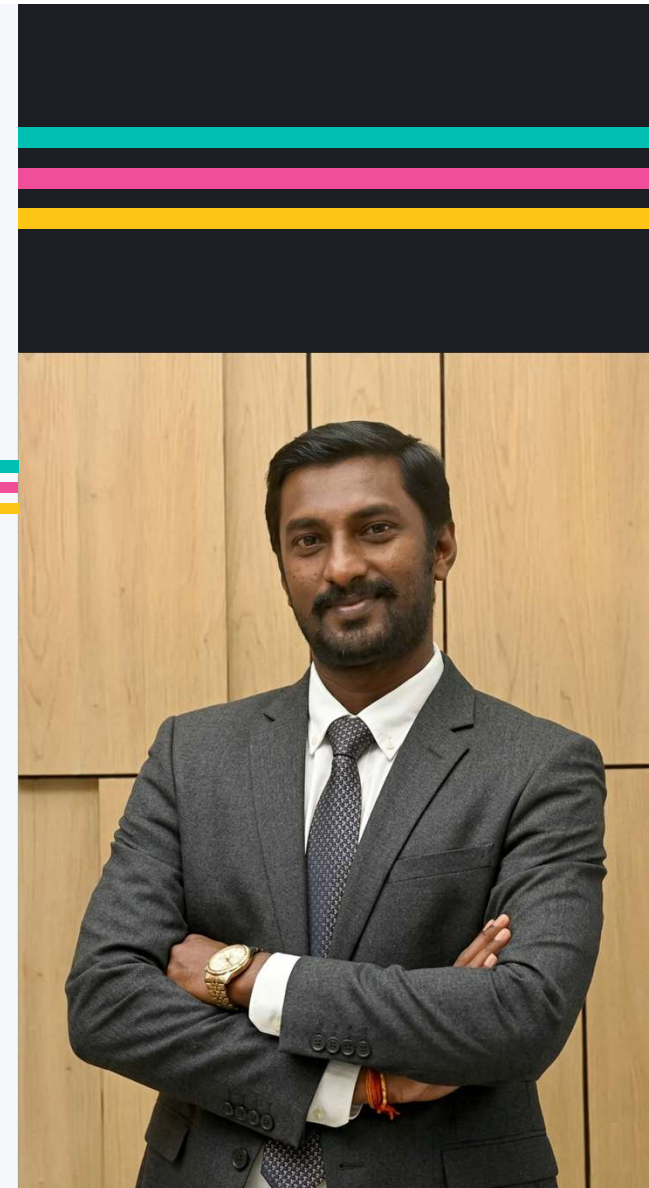




AI-Powered CloudTrail Threat Hunting with ESRE Semantic Search

Find the attack you didn't know to search for |



Cloud Trail

Records the walk

Every API call — console, CLI, SDK, service-to-service — captured as a discrete event.

01

Every step is named

userIdentity, eventTime, sourceIPAddress, eventName, requestParameters, errorCode.

03

Footprints stay put

Log-file validation, S3 Object Lock and org-wide trails keep the path tamper-evident.

02

The trail is the evidence

System of record for SOC 2, PCI-DSS, HIPAA and ISO 27001 access and change controls.

04

The Problem



There is nothing here



— THE APPROACH

What Elasticsearch adds —

ESRE

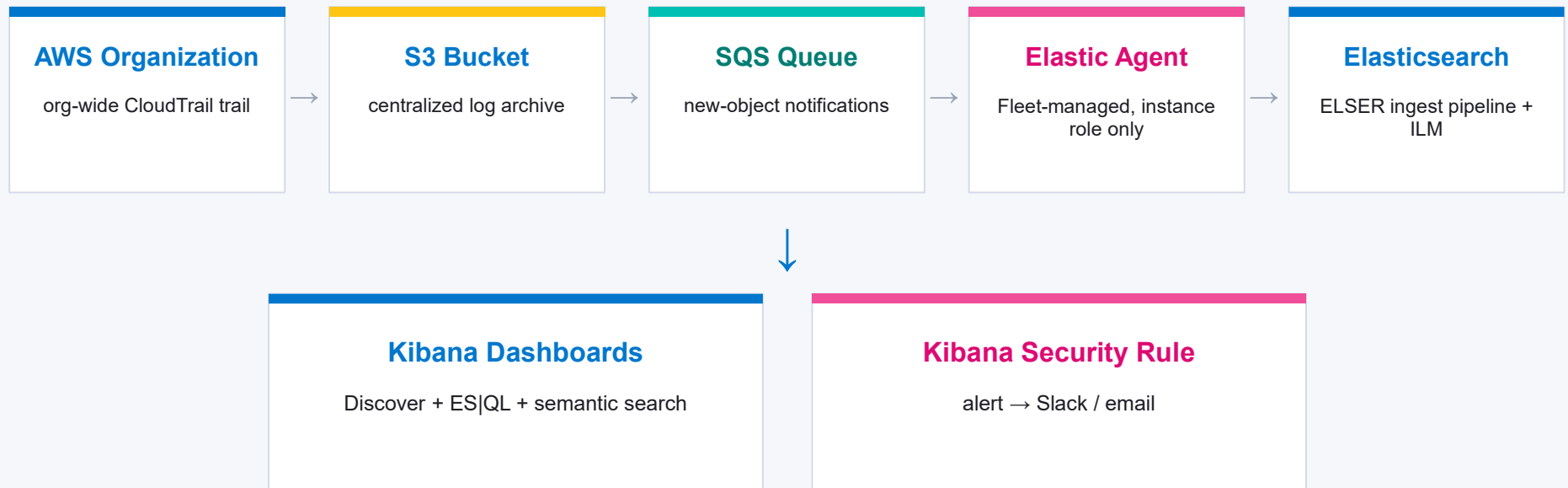
ES|QL

Fleet / Agent

Centralize → understand by meaning → analyze fast with ES|QL → alert automatically

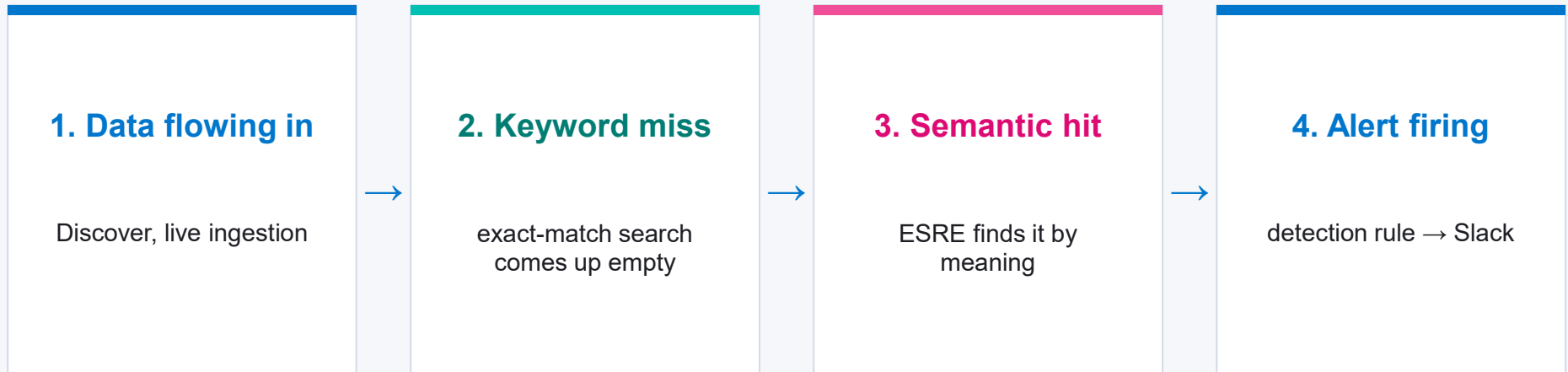
— ARCHITECTURE

End-to-end architecture



— LIVE DEMO

What you're about to see



— DEMO

Data flowing in

Discover — logs-aws.cloudtrail-*

@TIMESTAMP	EVENT_NAME	USER_IDENTITY.ARN	OUTCOME
07:42:11.203	AttachUserPolicy	arn:aws:iam::...:user/j.alvarez	success
07:41:58.771	ConsoleLogin	arn:aws:iam::...:user/j.alvarez	success
07:40:02.114	GetSecretValue	arn:aws:iam::...:role/ci-deploy	failure
07:38:47.590	CreatePolicyVersion	arn:aws:iam::...:user/j.alvarez	success
07:36:15.982	PutRolePolicy	arn:aws:iam::...:role/svc-billing	success

— DEMO

Keyword search vs. semantic search

KEYWORD SEARCH: “GIVE THEMSELVES MORE PERMISSIONS”

No results found

SEMANTIC SEARCH (ESRE) — SAME INTENT

AttachUserPolicy user/j.alvarez granted
iam:AdministratorAccess **0.94**

PutRolePolicy role/svc-billing given inline
wildcard policy **0.91**

CreatePolicyVersion new policy version
set as default **0.88**

— IMPACT

Results and impact

01 Faster investigation

02 Lower vocabulary bottleneck

03 Standing detection instead of manual, after-the-fact review

Let's Connect

- ❖ santhosh.n@sq1.security
- ❖ santhoshnarayanadass@gmail.com
- ❖ <https://www.linkedin.com/in/sndass/>



Santhosh NarayanaDass
CCSK | AWS & GCP Certified
Solutions Architect Professional |...

